Omkar Sudhir Patil

omkarpatil64328@gmail.com (352) 222-1713 omkar-patil-024 omkarsudhirpatil.com
patilomkarsudhir Google Scholar

Professional Summary

Researcher working at the intersection of controls, robotics, and AI to solve fundamental challenges in autonomous systems. Pioneered Lyapunov-based Deep Neural Networks (LbDNNs), solving a 25-year open problem by developing the first DNN-based controller with real-time adaptation and provable stability guarantees. Expertise spans adaptive control, multi-agent systems, and safety-critical applications, with research validated on real-world robotic platforms ranging from quadrotors to multi-robot coordination.

Education

University of Florida *Aug 2019 – May 2023*
Ph.D. in Mechanical Engineering (GPA: 3.97/4.0)

- **Advisor:** Dr. Warren E. Dixon
- **Dissertation:** "Implicit and Deep Learning-Based Control Methods for Uncertain Nonlinear Systems"
- **Research Focus:** Lyapunov-based deep neural networks, adaptive control theory, nonlinear control, RISE controllers, multi-agent systems, concurrent learning, physics-informed neural networks
- **Key Achievement:** Solved 25-year open problem extending Lyapunov-derived adaptive online update laws from shallow networks to deep neural networks of arbitrary depth

University of Florida *Aug 2019 – Aug 2022*
M.S. in Mechanical Engineering

- **Specialization:** Nonlinear control systems, adaptive control, robust control

Indian Institute of Technology Delhi *Jul 2014 – Aug 2018*
B.Tech in Production and Industrial Engineering

- **Research Area:** Control systems and robotics applications

Professional Experience

Postdoctoral Research Associate *May 2023 – Present*
University of Florida, Department of Mechanical & Aerospace Engineering

- **Research Leadership:** Lead development of next-generation adaptive control frameworks combining deep learning with mathematical rigor, securing $2M+ in research funding from ONR, AFOSR, and AFRL
- **Breakthrough Innovations:** Pioneered Lyapunov-based deep neural network (Lb-DNN) controllers with online weight adaptation, enabling real-time learning without pre-training requirements
- **Industry Applications:** Developed adaptive control solutions for aerospace systems, autonomous spacecraft servicing, quadrotor control, and multi-agent robotics with guaranteed stability
- **Publication Impact:** 20 papers in premier venues including IEEE TAC, IEEE T-NN, ACC, and CDC with 300+ citations
- **Mentorship:** Supervised 15 graduate students and postdocs, fostering collaborative research environment
- **Software Development:** Created MATLAB and Python packages for Lb-DNN controllers and projection algorithms, deployed on robotic platforms at UF Autonomy Park

Graduate Research Assistant *Aug 2019 – May 2023*
University of Florida, Nonlinear Controls and Robotics Group

- **Theoretical Breakthroughs:** Solved 17-year open question on exponential stability of RISE controllers, previously thought to achieve only asymptotic convergence

- **Algorithm Development:** Developed composite adaptive laws for deep neural networks using concurrent learning, achieving parameter convergence without persistent excitation
- **Multi-disciplinary Research:** Advanced control barrier functions, physics-informed neural networks, and stochastic control theory
- **Experimental Validation:** Implemented controllers on quadrotors, robotic manipulators, and multi-agent systems with real-time performance verification

Project Associate *May 2018 – Apr 2019*

Indian Institute of Technology Delhi, Department of Electrical Engineering

- **Aerial Robotics:** Designed and implemented hierarchical control systems for quadrotor aerial manipulators
- **Nonlinear Control:** Developed singularity-free control algorithms using barrier Lyapunov functions and saturation techniques
- **System Integration:** Programmed flight control systems and derived complete Lagrangian dynamics for aerial manipulation platforms

Key Research Contributions & Impact

Deep Learning & Control Theory Integration

- **Pioneering Framework:** First to extend Lyapunov-based adaptive control to deep neural networks of arbitrary depth, solving 25-year theoretical gap
- **Composite Adaptation:** Developed first composite adaptive laws for Lb-DNNs using concurrent learning, achieving parameter convergence guarantees
- **Real-time Learning:** Created online adaptation algorithms requiring no pre-training, enabling deployment in unknown dynamic environments

Fundamental Control Theory Advances

- **RISE Stability:** Proved exponential stability of RISE controllers, resolving 17-year open question in robust control
- **Stochastic Control:** Extended Lb-DNN framework to stochastic nonlinear systems with unknown drift and diffusion processes
- **Safety-Critical Systems:** Advanced adaptive control barrier functions using deep neural networks for real-time safety guarantee

Multi-Agent & Aerospace Applications

- **Spacecraft Control:** Developed distributed controllers for collaborative spacecraft servicing under partial state feedback
- **Target Tracking:** Created exponential convergence algorithms for heterogeneous multi-agent systems with limited communication
- **Herding Control:** Designed adaptive indirect herding strategies for multiple target agents with unknown interaction dynamics

Technical Skills

Control Theory & Mathematics: Lyapunov stability analysis, adaptive control, nonlinear control, robust control, optimal control, stochastic control, multi-agent systems, differential geometry, functional analysis, convex optimization

Machine Learning & AI: Deep neural networks, physics-informed neural networks, reinforcement learning, concurrent learning, online learning algorithms, ResNet architectures, LSTM networks, graph neural networks

Programming & Software: MATLAB/Simulink, Python, C++, ROS, Git, LaTeX, TensorFlow, PyTorch, control system toolboxes, numerical optimization solvers

Experimental Platforms: Quadrotor UAVs, robotic manipulators, multi-agent testbeds, real-time control systems, embedded systems, hardware-in-the-loop simulation

Interactive Research Demonstrations

- **LbDNN Quadrotor Control:** 3D interactive simulation demonstrating real-time neural network adaptation with feedback loss
- **CBF Lab:** Educational platform for Control Barrier Functions with safety-critical control demonstrations
- **Physics-Informed Neural Networks:** Interactive demos for scientific machine learning with automatic differentiation
- **Neural ODE Visualizer:** Real-time continuous-time deep learning with dynamic system modeling
- **SLAM Tutorial:** Interactive robotics localization and mapping educational tool

Selected Publications

Book

- O. Patil, E. Griffis, and W. E. Dixon, *Deep Learning-Based Adaptive Control: A Lyapunov-Based Approach*. In development with Springer, 2025.

Journal Papers

- O. Patil, R. Kamalapurkar, and W. Dixon, ”A saturated RISE controller with exponential stability guarantees,” *IEEE Trans. Autom. Control*, 2025.
- O. Patil, D. Le, E. Griffis, and W. Dixon, ”Lyapunov-based deep residual neural network (ResNet) adaptive control,” *IEEE Access*, 2025.
- W. Makumi, O. Patil, and W. Dixon, ”Lyapunov-based adaptive deep learning for approximate dynamic programming,” *Automatica*, vol. 180, p. 112462, 2025.
- O. Patil, R. Sun, S. Bhasin, and W. E. Dixon, ”Adaptive control of time-varying parameter systems with asymptotic tracking,” *IEEE Trans. Autom. Control*, vol. 67, no. 9, pp. 4809–4815, 2022.
- O. Patil, D. Le, M. Greene, and W. E. Dixon, ”Lyapunov-derived control and adaptive update laws for inner and outer layer weights of a deep neural network,” *IEEE Control Syst Lett.*, vol. 6, pp. 1855–1860, 2022.

Awards

- **Graduate Student Research Award**, 2023, for outstanding research as a graduate student in the Department of Mechanical and Aerospace Engineering, University of Florida
- **BOSS Award**, 2018, for the best hardcore experimental project in Mechanical Engineering discipline, IIT Delhi

Invited Talks

- “Lyapunov-derived control and adaptive update laws for inner and outer layer weights of a deep neural network”, *Resilient and Autonomous Systems Lab (RASLab), Florida State University*, Dec. 2021
- “Deep Residual Neural Network (ResNet)-based Adaptive Control: A Lyapunov-based Approach”, *AFOSR COE Program Review, University of Florida*, April. 2023
- “Composite Adaptive Lyapunov-based Deep Neural Network Control”, *AFOSR COE Program Review, Duke University*, Dec. 2023

Teaching Experience

- Teaching Assistant (UF), Vibrations, Spring 2020

Peer Review Service

- IEEE Transactions on Automatic Control
- Automatica
- IEEE Control Systems Letters
- IEEE Transactions on Control Systems Technology
- IEEE/ASME Transactions on Mechatronics
- IEEE Transactions on Neural Networks and Learning Systems
- International Journal of Adaptive Control and Signal Processing
- Advances on Neural Information Processing Systems (NeurIPS)
- International Conference on Learning and Representations (ICLR)
- IEEE Conference on Decision and Control
- American Control Conference
- IFAC World Congress